

GEI Consultants



Geologic Interpretive Summary Report

GRCA 318719: Rehabilitate North Rim Utility Infrastructure

Waterline Construction from the North Rim Ballfield Site to Roaring Springs Pump House

Grand Canyon North Rim, Arizona

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March 31, 2026
Project No. 2305987





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March 31, 2026
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VIA EMAIL: matt.gough@hdrinc.com

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Re: Geologic Interpretive Summary Report
Waterline Construction from the North Rim Ballfield Site to Roaring Springs Pump House
GRCA 318719: Rehabilitate North Rim Utility Infrastructure
Grand Canyon North Rim, Arizona

Mr. Gough:

Submitted herewith is the Geologic Interpretive Summary Report for the Waterline Construction from the North Rim Ballfield Site to the Roaring Springs Pump House, a component of the GRCA 318719 Rehabilitate North Rim Utility Infrastructure project. This report was prepared in general accordance with the Agreement between GEI Consultants, Inc. and HDR Engineering, Inc. The Geologic Interpretive Summary Report includes discussions of the geological conditions, geotechnical data, and interpretations for the subsurface conditions that are anticipated to be encountered during the proposed construction, and construction considerations in support of the subject project.

If you have any questions regarding the contents of this report, please contact the undersigned.

Sincerely,

GEI Consultants, Inc.

Rob D. Johanson, P.E.
Project Engineer

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Table of Contents

Report Organization iv

1. Introduction 1
1.1. Project Description 1
1.2. Description and Purpose of the GISR 1

2. Project Background 3
2.1. Project Overview 3
2.1.1. Existing System 3
2.1.2. Proposed Improvements 3
2.2. Subsurface Investigations 4
2.2.1. North Rim Investigation 4
2.2.2. Inner Canyon Investigation 5
2.2.3. Geophysical Investigation 5
2.3. Literature Review 6
2.4. Historic Projects 6
2.4.1. Transcanyon Water Transmission System Improvements Project 6
2.4.2. Offset Wells 8

3. Geology 10
3.1. Regional Geology 10
3.2. Site Geology 11
3.2.1. Kaibab Formation 11
3.2.2. Toroweap Formation 12
3.2.3. Coconino Sandstone 13
3.2.4. Hermit Formation 14
3.2.5. Supai Group 14
3.2.6. Redwall Limestone 16
3.2.7. Temple Butte Formation 17
3.2.8. Tonto Group 18

4. Anticipated Geologic Conditions and Ground Behavior 20
4.1. General 20
4.2. Geologic Formation Characteristics 20
4.2.1. Kaibab Formation 20
4.2.2. Toroweap Formation 21
4.2.3. Coconino Sandstone 22
4.2.4. Hermit Formation 22

4.2.5. Supai Group 23
4.2.6. Redwall Limestone 24
4.2.7. Temple Butte Formation 26
4.2.8. Tonto Group 26
4.3. Rock Properties 27
4.4. Anticipated Hydrogeologic Conditions 28
4.4.1. Groundwater 28
4.4.2. Hydraulic Conductivity and Cement Take 28

5. Construction Considerations 30
5.1. General 30
5.2. Anticipated Drilling Conditions 30
5.3. Casing Pipe Design 31
5.4. Drill Bit Considerations 32
5.5. Cementing 32

6. Limitations 34

7. References 35

List of Tables

Table 1-1. Project Roles 1
Table 2-1. South Rim Bore Problems and Solutions 8
Table 4-1. Estimated Rock Type Percent by Volume of Fossil Mountain Member 21
Table 4-2. Estimated Rock Type Percent by Volume of Woods Ranch Member 21
Table 4-3. Estimated Rock Type Percent by Volume of Brady Canyon Member 21
Table 4-4. Estimated Rock Type Percent by Volume of Seligman Member 22
Table 4-5. Estimated Rock Type Percent by Volume of Coconino Sandstone 22
Table 4-6. Estimated Rock Type Percent by Volume of Hermit Formation 22
Table 4-7. Estimated Rock Type Percent by Volume of Esplanade Sandstone 23
Table 4-8. Estimated Rock Type Percent by Volume of Wescogame Formation 23
Table 4-9. Estimated Rock Type Percent by Volume of Manakacha Formation 24
Table 4-10. Estimated Rock Type Percent by Volume of Watahomigi Formation 24
Table 4-11. Estimated Rock Type Percent by Volume of Horseshoe Mesa Member 25
Table 4-12. Estimated Rock Type Percent by Volume of Mooney Falls Member 25
Table 4-13. Estimated Rock Type Percent by Volume of Thunder Springs Member 25
Table 4-14. Estimated Rock Type Percent by Volume of Whitmore Wash Member 26
Table 4-15. Estimated Rock Type Percent by Volume of Temple Butte Formation 26
Table 4-16. Estimated Rock Type Percent by Volume of Muav Formation 26
Table 4-17. Estimated Rock Type Percent by Volume of Bright Angel Formation 27
Table 4-18. Estimated Strength and Abrasivity Parameters by Rock Type 27
Table 4-19. Estimated Hydraulic Properties 29
Table 5-1. Anticipated Drilling Conditions 31

Table 5-2. Estimated Cement Take by Formation 32

List of Figures

Figure 2-1. Site Vicinity and Layout 9
Figure 5-1. Interpreted Geologic Profile 33

Appendices

Appendix A Tunnelman's Ground Classification Chart

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Report Organization

The Geologic Interpretive Summary Report (GISR) is organized as follows:

Section 1: Introduction

Includes the project description and discusses the purpose and intended usage of the GISR.

Section 2: Project Background

Provides an overview of the project in relation to the proposed construction, a description of the geologic and subsurface investigations undertaken to evaluate ground conditions at the site, and a summary of relevant historic projects.

Section 3: Geology

Provides an in-depth description of the regional and site-specific geology in the project site vicinity.

Section 4: Anticipated Geologic Conditions and Ground Behavior

Provides anticipated formation characteristics and material properties of each rock type. Additional properties are provided for hydrogeological conditions.

Section 5: Construction Considerations

Briefly discusses relevant construction considerations for anticipated drilling conditions, casing pipe, drill bits, and cementing.

Sections 6 and 7: Limitations and References

Describes limitations of the GISR and provides references for citations within the report.

1. Introduction

1.1. Project Description

The National Park Service (NPS) is currently planning infrastructure improvements to the existing water and wastewater systems for its North Rim and Inner Canyon facilities located in Grand Canyon National Park, Arizona as part of the GRCA 318719: Rehabilitate North Rim Utility Infrastructure Project (Project). The Project includes the construction of a new water treatment plant on the North Rim, upgrades to the existing wastewater treatment plant, open cut pipeline construction, and the replacement of the existing North Rim Waterline between North Rim and the Roaring Springs Pump House (RSPH) using directional drilling methods. Table 1-1 below lists organizations and their respective roles for the Project.

Table 1-1. Project Roles

Organization	Role
National Parks Service (NPS)	Owner
HDR Engineering, Inc. (HDR)	Prime Engineer
GEI Consultants, Inc. (GEI)	Geotechnical and Trenchless Design Engineer
Geostock Sandia (GKS)	Directional Drilling Consulting Engineer
Collier Geophysics (Collier)	Geophysical Investigations

This report summarizes ground conditions that are anticipated to be encountered during directional drilling required to construct the North Rim Waterline from the North Rim to the RSPH. For the purposes of this report, directional drilling refers to the drilling practice commonly used in the oil and gas industry in which a hole is drilled through soil and rock with a vertical entry with the ability to steer the drill string using various bottom hole assembly configurations. A summary of the report organization is provided at the beginning of this report. Section 1.2 provides the description and purpose of this Geologic Interpretive Summary Report (GISR).

1.2. Description and Purpose of the GISR

The main purpose of this GISR is to summarize interpreted ground conditions and behavior that are anticipated to be encountered during directional drilling between the entry point on the North Rim and the exit location at the RSPH. Geological and subsurface conditions observed through site visits, geotechnical and geophysical field explorations, laboratory analyses of collected samples, and literature review are summarized herein and presented for informational purposes only.

The interpretations presented in this GISR for the anticipated subsurface conditions may be used for informational purposes and to help the Contractor evaluate allowed means and methods. This GISR also summarizes geological, geotechnical, and construction considerations that form the basis of the directional drilling portion of the Project design and Contract Documents. The GISR is prepared in tandem with the applicable Contract Documents.

The interpreted subsurface conditions presented in this GISR reflect the design team's judgment of anticipated subsurface conditions and ground behavior based on assumed construction means and

methods expected to be employed in the work. Interpretations were established by considering available geological and geotechnical data, and previous construction experience in similar subsurface conditions. Development of the Project design required interpretation, extrapolation, and interpolation between data obtained for the Project including aerial imagery, geophysical profiles, geologic mapping, geotechnical test boreholes (i.e., borings), and laboratory tests. This report has been prepared exclusively for our client for design, bidding, and construction planning for the subject project. GEI is not responsible for technical interpretations by others of the data presented in this report or use of this report by others for the subject project or other projects. The interpreted geology and estimated values presented in this report are only intended as supplemental information for the proposed design and construction as understood by GEI at the time of issuing this report.

2. Project Background

2.1. Project Overview

2.1.1. Existing System

The North Rim waterline was originally constructed in 1928 to supply potable water to the Grand Canyon Lodge on the North Rim. The existing waterline conveys water from the Roaring Springs Cave to the North Rim by means of a pump house located below Roaring Springs and several sections of surface mounted pipelines. Water exiting the Roaring Springs cave is conveyed through a 12-inch steel pipe from the cave entrance to an existing diversion/spill box followed by approximately 700-feet of aluminum pipe where it is fed by gravity to the pump house. Raw water from the Roaring Springs Cave is treated at the pump house and pumped through an approximately 12,500-foot long 3.5-inch steel pipeline to the North Rim water storage tanks. Treated water stored in the North Rim water tanks is distributed throughout the facilities located on the North Rim.

The existing, surface mounted pipeline is susceptible to environmental hazards including rockfall, treefall, and freeze/thaw cycles. These hazards in combination with the age of the pipeline result in frequent damage to the pipeline. Repairs to the pipeline are dangerous, challenging, and costly due to the remote location and steep, rough terrain along the pipeline. Additionally, the susceptibility to freezing temperatures requires the park service to drain the pipeline in the winter months during which the park must operate exclusively on water stored within the North Rim water tanks.

2.1.2. Proposed Improvements

Proposed improvements include upgrades to the existing RSPH and replacement of the existing North Rim waterline between the RSPH and North Rim Water Tanks with 4.5-inch raw and finished waterlines, The raw waterline will bring water from the RSPH to the North Rim for treatment, while the finished waterline will return potable water to the RSPH and other Inner Canyon facilities.

The waterlines are planned to be constructed in three segments including the following as shown in Figure 2-1:

1. Segment 1 - Roaring Springs Cave to the RSPH
2. Segment 2 - RSPH to the North Rim Ballfield Site
3. Segment 3 - North Rim Ballfield Site to the North Rim Water Tank

This GISR addresses only the second segment of the alignment between the RSPH and the North Rim Ballfield Site as described above. This segment will include the 4.5-inch raw waterline and the 4.5-inch finished waterline, installed using directional drilling methods in separate cased bores which will parallel each other and enter/exit the ground near each other. Therefore, the anticipated geologic conditions with depth at each bore are assumed to effectively be the same. The bores will each include identical telescoped casing, excluding the final casing, which differs slightly between the two bores. Telescoped casings include an 80-foot long 20-inch diameter conductor casing, 1,350-foot long 13 3/8-inch diameter

surface casing, 8,800-foot long 9 5/8 -inch diameter intermediate casing, and a 9,293 foot long 7-inch diameter final casing for bore B and a 9,304 foot long 7-inch diameter final casing for bore A. The bores will have vertical lengths or true vertical depths (TVD) of approximately 3,314 feet for bore B and 3,319 feet for bore A, horizontal lengths or displacements of approximately 7,200 feet for both bores, total measured depths along the bore path (MD) of approximately 9,293 feet for bore B and 9,304 feet for bore A, and will both exit the side of the canyon at a near horizontal angle.

Directional drilling was determined to be the preferred construction method during the cost and scope validation phase due to the success of a previous and similar bore conducted at the South Rim of the Grand Canyon, reduced construction costs, reduced hazards to construction workers, and reduced environmental impact in comparison to other alternatives. Horizontal Directional Drilling (HDD), often used in municipal or near surface pipeline construction, is not feasible due to the length of the bore, major vertical relief of the canyon walls, and limited equipment staging and access to the lower elevation Inner Canyon side of the alignment.

2.2. Subsurface Investigations

2.2.1. North Rim Investigation

GEI performed a geotechnical subsurface investigation at the North Rim for the proposed water treatment plant, improvements to the existing wastewater treatment plant, and raw water pipeline alignment between October 2022 and November 2023. The first phase of investigation was completed in October 2022 and included percolation testing and a test pit investigation at the existing wastewater treatment plant. The second phase of investigation was conducted in November 2022 and consisted of 17 geotechnical borings drilled at the existing wastewater treatment plant, proposed water treatment plant, proposed raw water pipeline alignment between the North Rim Water Tanks and North Rim Ballfield Site, and proposed bore entry locations. The third phase of investigation was completed in November 2023 and consisted of 15 additional geotechnical borings at the existing wastewater treatment plant, proposed water treatment plant, and proposed raw water pipeline between the North Rim Water Tanks and North Rim Ballfield Site. During the second and third phases of investigation, GEI collected split spoon and Modified California samples of the onsite residual soils and HQ core samples of the Kaibab Formation bedrock. A geotechnical laboratory testing program was performed on representative samples of the soil and bedrock and the results assisted in classification and assessment of applicable engineering properties for encountered subsurface materials.

Details of the procedures used for conducting the field work and laboratory testing are summarized in the North Rim Geotechnical Investigation Report (GIR) (GEI, 2025). Though the primary purpose of the North Rim investigation was to characterize the subsurface conditions for the proposed North Rim water and wastewater treatment facility improvements, core samples obtained during the investigation were used for determination of Kaibab Formation properties included herein. Geotechnical data collected during the North Rim investigation relevant to directional drilling are summarized within this GISR.

2.2.2. Inner Canyon Investigation

GEI and GKS performed a geological and geotechnical field investigation for the directional drilling design and proposed Inner Canyon infrastructure improvements in November 2023. The investigation consisted of visual rock and soil classification along the North Kaibab Trail, hand excavation of test pits, in situ rock strength tests, and visual observation of the existing Inner Canyon infrastructure. During the investigation along the North Kaibab Trail, GEI and GKS collected rock samples, performed Schmidt Hammer rebound testing, and noted geologic formational contact elevations as observed hiking along the North Kaibab Trail. Geotechnical laboratory testing and a thin section analysis were performed by Metarock Laboratories on rock samples collected during the Inner Canyon investigation. The laboratory testing program was developed by GKS with input from GEI.

Details of the procedures used for conducting the field work and laboratory testing are summarized in the Inner Canyon GIR (GEI, 2025) and the North Rim Bore Drilling Plan (GKS, 2026). The Inner Canyon GIR focuses on the proposed improvements to the Inner Canyon facilities while the North Rim Drilling Plan provides an indicative drilling plan for directional drilling based on the results of the field investigation. Geotechnical data collected as part of the Inner Canyon investigation relevant to directional drilling are summarized within this GISR. All bedrock testing was performed on samples collected from the surface and were therefore subject to weathering. Weathering of surface rocks typically results in lower rock strength. As such, rock strength was conservatively estimated based on literature values for each lithology, with bedrock testing used for approximative confirmation of selected values.

2.2.3. Geophysical Investigation

Collier was retained by GEI to scope and design a multi-phase geophysical investigation to help characterize the subsurface geology for the proposed water treatment plant, bore entry locations, and bore exit locations, and to inform the design and completion of the water transmission bores. The geophysical survey was completed in two phases which included the “shallow phase” Electrical Resistivity Tomography (ERT) survey for imaging the upper 100-200 feet below ground surface and the “deep phase” Seismic Reflection survey for imaging depths ranging from 200 feet to over 3,500 feet below ground surface.

Descriptions of the procedures and results for the geophysical investigation are presented in the North Rim Geophysical Studies Summary Report and Appendices (GEI, Collier, GKS 2025). Relevant findings of the geophysical investigation are accounted for in the geologic and hydrogeologic interpretations summarized in Section 4. The ERT methods used in the geophysical investigation are limited to depths of approximately 100-feet to 200-feet and are only indicative of relatively near surface formations, while 3D seismic reflection methods were employed to image depths to 3,500 feet below ground surface. The purpose of the additional geophysical investigations was to confirm the presence and approximate location of contacts between geologic formations, identify the presence of groundwater and preferential flow pathways, and characterize potential karst features, fractures, and anomalous rock zones at greater depths along the bore path.

2.3. Literature Review

In addition to the subsurface investigation, GEI performed a review of available literature including available Geologic Maps and LiDAR data from the United States Geological Survey (USGS) and other publications pertaining to geological and hydrogeological conditions at the Grand Canyon. Specific documents reviewed for relevant data include:

- Aerial imagery through Google Earth. Images dating from the 1980s through present were reviewed and showed little evidence of recent surficial geologic activity or observable disturbance of exposed geologic units near the North Rim bore site since the earliest available imagery.
- Available literature from the South Rim bore including feasibility study from McLaughlin Water Engineers (1984), hydrogeologic study from Hargis and Montgomery, Inc. (1984), and summary article from Lattimore et al. (1987). Relevant information from the South Rim bore is summarized in Section 2.4.1.
- Available USGS Geologic Maps covering the bore path area. Maps were used to identify the bedrock formations and structural features which aided in characterizing the rock mass and identifying geologic contacts.
- Available USGS LiDAR imagery of the Grand Canyon. LiDAR imagery with resolution less than one meter was used to identify surface weathering patterns that are indicative of changes in lithology for the determination of geologic contact elevations.
- Available literature pertaining to rock strength and hydraulic characteristics by rock type including Atwell and Farmer (1976) and Freeze and Cherry (1979).
- Geophysical CSAMT survey by Zonge International (2015) and hydrogeologic study by Yeh and Associates (2015).
- The geologic interpretations presented in this report are derived from USGS geologic unit descriptions, literature from published books and academic papers, as well as unpublished literature from the South Rim Bore.

Relevant observations from the literature review are noted in the GISR text where findings were applicable, and geologic interpretations were developed accordingly.

2.4. Historic Projects

2.4.1. Transcanyon Water Transmission System Improvements Project

Located approximately ten miles south of the North Rim bore site at the South Rim of the Grand Canyon, the Transcanyon Water Transmission System Improvements project is the only other known project to have included the installation of a utility pipeline by means of directional drilling. As such, GEI has relied on data, observations, direct involvement from participants of the project, and design elements from this project in developing this GISR.

The Transcanyon Water Transmission System Improvements project was originally conceived in 1983 following an NPS study indicating that the demand for water at the South Rim was rapidly outpacing the capacity of the existing South Rim water supply system. Despite the relatively young age of the existing system, the NPS proposed to add a second transmission pipeline from Havasupai Gardens (then called Indian Gardens) to the South Rim to increase water supply. A 1984 feasibility study was completed to assess the feasibility of directional drilling methods to install the new pipeline (McLaughlin Water Engineers, 1984). The report concluded that directional drilling was preferable to alternative construction methods due to lower cost, reduced environmental impact, and fewer hazards to construction workers. The feasibility report noted that risks associated with directional drilling could have serious impacts on the project, but that those same risks are associated with alternative construction methods. With the directional drilling alternative these risks were considered to have a lower probability of occurrence and could be mitigated with careful drilling management and operation.

Drilling operations to construct the waterline were summarized in an SPE/IADC paper written following completion of the South Rim bore (Lattimore et al. 1987). Directional drilling was completed in 1986 and required 49 days to drill from spud (entry) to the total depth of 5,075 feet. Drilling mud was used to drill the conductor hole to 80 feet and the surface hole to 250 feet. No circulation loss of the drilling mud was encountered while drilling the conductor hole and the surface hole to the 250-feet depth. Due to the proximity to the cliff face and the risk of surface contamination, drilling mud was not considered a viable circulation medium for the remainder of the bore. Rather, drilling was completed using air and mist to circulate cuttings to the surface. However, circulation was lost with the mist and only intermittently returned while drilling the rest of the South Rim bore. Much of the circulation losses were attributed to regional northeast- and northwest-trending fracture systems and solution cavities in the limestone formations. During drilling, directional measurements were completed using magnetic single-shot instruments and confirmed with several comparison gyro runs. The bore was completed using three runs of casing pipe telescopically installed to progressively greater bore depths. Installed casing pipe included a 66-foot section of 26-inch conductor casing, a 245-foot section of 20-inch surface casing, and a 10 3/4-inch casing run to exit the canyon wall. The 8 5/8-inch steel waterline was run through the final casing before releasing the hole to the NPS for connection to the Havasupai Gardens pump station.

Geology encountered during directional drilling consisted of a Paleozoic sedimentary sequence from Permian to Cambrian age. Geologic formations from oldest to youngest included the Muav Formation, Temple Butte Formation, Redwall Limestone, Supai Group, Hermit Formation, Coconino Sandstone, Toroweap Formation, and Kaibab Formation. No major faults or other structural features were encountered during drilling and all formation top elevations as projected from outcrops at the South Rim were consistent with a dip of approximately one to two degrees to the southwest as documented in USGS geologic maps. Several of the limestone units exhibited karst features such as dissolution cavities and voids, particularly in the Kaibab, Toroweap, and Redwall Limestone Formations.

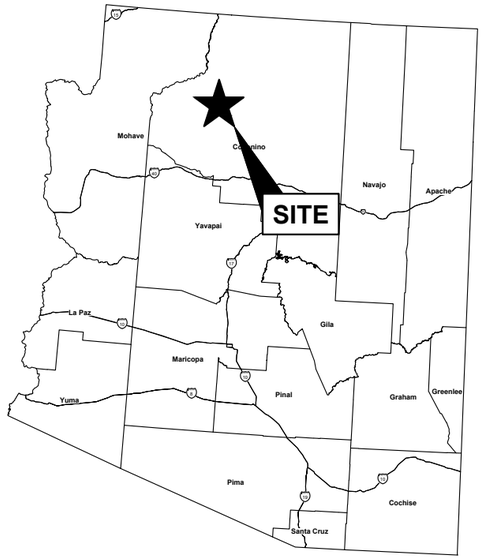
Several issues were encountered during the drilling process. Table 2-1 describes the major problems encountered during drilling and the solutions employed in the field.

Table 2-1. South Rim Bore Problems and Solutions

Problem	Solution
Severe drill string damage/loss of Bottom Hole Assembly (BHA) components	Fishing and/or cementing section of hole and sidetracking
Loss of air and mist circulation due to fractures and dissolution cavities	Continue drilling, monitor torque and drag to infer degree of hole cleaning
Anticipated difficulty setting casing due to severe doglegs resulting from sidetracks and drilling through dissolution cavities	Preemptive lubrication of hole using polymer, foam, and air was used but unknown whether this was necessary
Hole cleaning issues causing the hole to become sticky near the end	The motor run was aborted and the last directional correction was not completed causing the directional bore to exit the canyon wall in a slightly different location
Cement losses to formation due to fractures and dissolution cavities during cementing	Field testing developed to ensure fluids were able to travel through annular space between the casing and bore wall from top to bottom

2.4.2. Offset Wells

Other comparable wells completed in the vicinity of the North Rim were drilled for the purposes of oil and gas exploration and are located over 30 miles north of the project site. Additional details regarding these offset wells are provided in North Rim Drilling Plan (GKS, 2026). Large variation between geologic formation composition and thickness have been observed over these distances. Therefore, data from offset wells was not used in the interpretations summarized within this report.

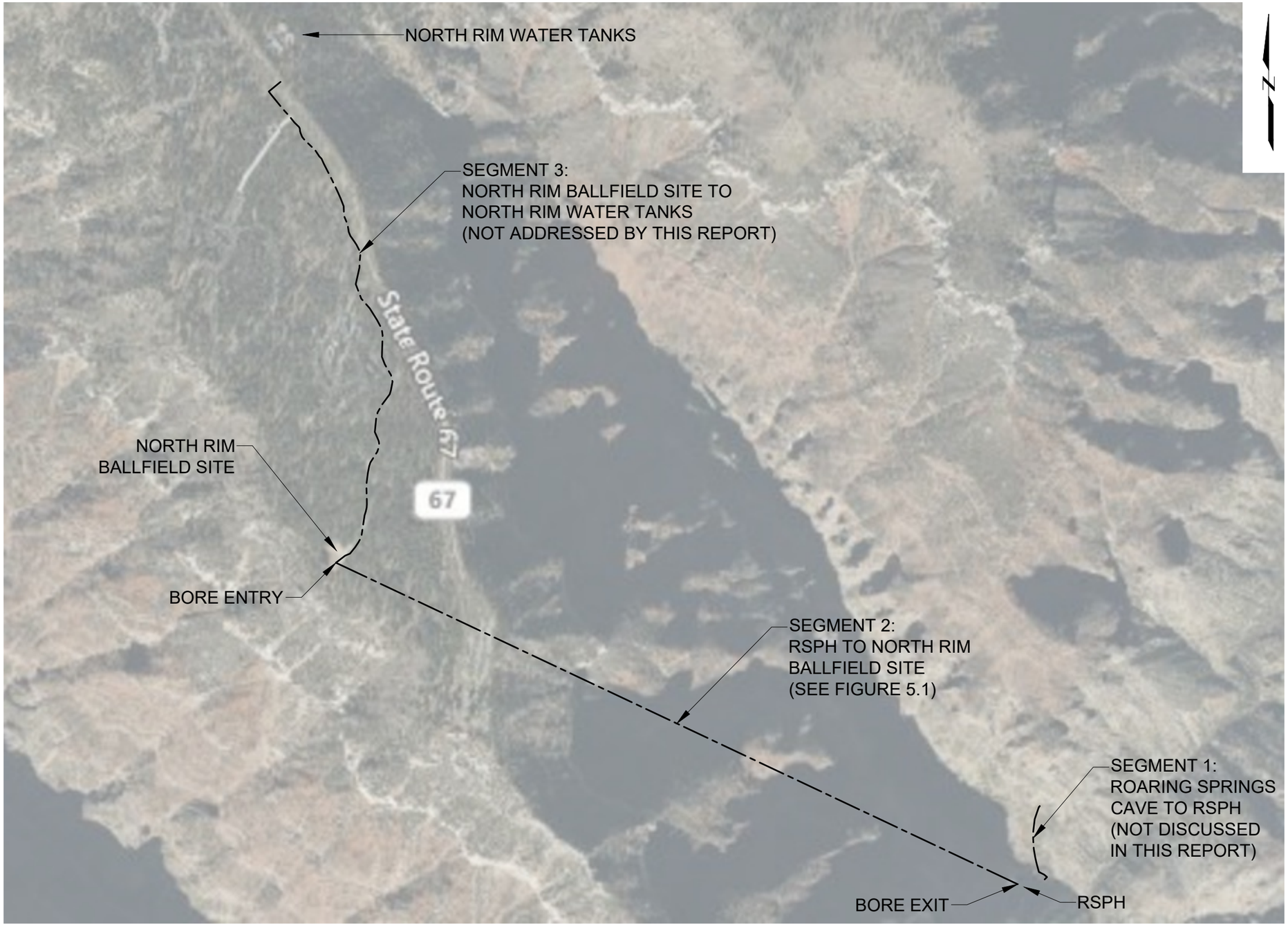


LEGEND:

PROPOSED PIPELINE ALIGNMENTS

NOTES:

1. ALL LOCATIONS ARE APPROXIMATE. SEE CONTRACT DRAWINGS FOR EXACT LOCATIONS.
2. ANTICIPATED GEOLOGIC PROFILE FOR BORE ALIGNMENT SHOWN IN FIGURE 5-1.



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GRCA 318719 Rehabilitate North Rim Utility Infrastructure
Grand Canyon North Rim, Arizona

HDR Engineering, Inc.
Fort Collins, Colorado

GEI Consultants

SITE VICINITY AND LAYOUT

Project 2305987 | July 2025 | Fig. 2-1

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3. Geology

3.1. Regional Geology

The Grand Canyon is located in the southwestern corner of the Colorado Plateau Physiographic Province, which spans the four corners region of Arizona, New Mexico, Colorado, and Utah. The Colorado Plateau is a region of thick continental crust which has remained largely resistant to significant tectonic deformation since the end of the Precambrian (544 million years before present (Ma)). A 325-million-year period of net subsidence and deposition during the Paleozoic was followed by Laramide-era uplift 75-55 Ma, and subsequent downcutting of the canyon in the last 5 million years (Huntoon, 2003). This has resulted in the prominently exposed geologic units within the present-day Grand Canyon which are both well preserved and readily observable.

The sedimentary units which form the Grand Canyon are founded on Precambrian basement rocks. In the eastern portion of the Grand Canyon sedimentary rocks of the Cambrian Tonto Group (Tapeats Sandstone, Muav Formation, and Bright Angel Formation) rest on tilted beds of the Proterozoic (1.4- to 1.1-billion-year-old) Grand Canyon Supergroup, along the “Great Unconformity”, which represents an approximate 600-million-year gap in time of the geologic record at this location. The pre-existing topography formed by the Precambrian rocks which the Tonto group is founded on included irregular rolling hills, with alternating peaks and valleys of resistant bedrock separated by intermediate lowlands. The Tonto Group constitutes the foundation of the Paleozoic stratigraphic sequence forming the canyon walls, which are typically capped by the Permian Kaibab Formation (Beus and Morales, 2003).

Major faults in the region are of Precambrian age (>544 Ma), generally trend northeast, north, and northwest, and have undergone periodic reactivation from cycles of regional compressive and tensional tectonic stresses (Huntoon, 2003). Faults near the site mapped on a 1:100,000 scale USGS quadrangle include the Roaring Springs Fault and the Bright Angel Fault (USGS, 2000). The bore entry point is located approximately 0.6 miles southwest of Roaring Springs Fault which is a reverse fault trending northwest-southeast along Roaring Springs Canyon. Based on the USGS quadrangle, the proposed bore alignment may cross the Roaring Springs Fault approximately 1.2 miles southeast of the entry point, within the Bright Angel Formation (USGS, 2000). The precise location of the Roaring Springs Fault is difficult to locate in the area in which the bore is anticipated to cross it per the USGS geologic map. The Roaring Springs Fault terminates in the Bright Angel Fault approximately 1.7 mi southeast of the bore entry point. Although it is not anticipated to cross the proposed bore path, the Bright Angel Fault is a major northeast-trending structural feature which crosscuts the Grand Canyon and forms the primary structural control for Bright Angel Creek.

Throughout the Paleozoic and Mesozoic era, the Grand Canyon has generally not been subject to significant brittle deformation from faulting, but has instead undergone broad, ductile deformation and variable cycles of regional subsidence and uplift (Huntoon, 2003). Deformation since the Precambrian has typically been marked by reactivation of the pre-existing faults, resulting in the formation of regional monoclines during the Laramide Uplift. The Kaibab Plateau where the site is located is one such Laramide-era monocline, thrusted upward from regional compressive stresses and reactivation of northwest-trending Precambrian faults (Huntoon, 2003). Because movement of crustal blocks has

typically occurred along major pre-existing Precambrian faults, smaller scale secondary faulting in the vicinity of the site is not anticipated to be encountered.

3.2. Site Geology

The proposed bore entry point is in the southern portion of the Kaibab Plateau along the North Rim of the Grand Canyon, directly south of the North Rim Campground. A 1:100,000 scale USGS quadrangle published in 2000 titled, Geologic Map of the Grand Canyon 30' x 60' Quadrangle (USGS, 2000) was used to determine occurrence and aerial coverage of major stratigraphic units anticipated to be encountered along the proposed bore path. These include (in descending stratigraphic order): Kaibab Formation, Toroweap Formation, Coconino Sandstone, Hermit Formation, Supai Group Units, Redwall Limestone, Temple Butte Formation, Muav Formation, and Bright Angel Formation. Though not anticipated to be encountered along the proposed bore path, the Tapeats Sandstone is also present on the USGS geologic map and may outcrop near the bore exit location. A supplementary pamphlet provided with the USGS map also provides summary descriptions of each unit, which are referenced below. The geology described here is generally applicable to the Kaibab Plateau region of the Grand Canyon. A cross section illustrating the geologic profile interpretation along the bore path is provided in Figure 5-1.

3.2.1. Kaibab Formation

The Lower Permian Kaibab Formation is a light tan to dark gray limestone, with variable bedding thickness (Hargis & Montgomery, 1984). It is a sandy dolomitic limestone, heavily jointed, and contains chert. The Kaibab was reported by Hargis & Montgomery to range in total thickness from about 250 to 300 feet near the historic south rim bore location.

The surface of the Kaibab Plateau and much of the Grand Canyon is capped by the Kaibab Formation. Generally, it is composed of both carbonate and siliciclastic sediments deposited throughout a complex cycle of marine transgressions and regressions (Hopkins and Thompson, 2003).

The Kaibab Formation is identifiable as gray cliffs formed over the vegetated slope of the underlying Toroweap Formation. Based on an isopach map provided in Hopkins and Thompson (2003), the Kaibab is between 300 to 400 feet thick throughout the Kaibab Plateau. It includes the upper Harrisburg member and the underlying Fossil Mountain member, separated by a distinct chert horizon. Based on the geologic investigation, only the Fossil Mountain Member of the Kaibab Formation will be encountered in the bore.

3.2.1.1. Fossil Mountain Member

The Fossil Mountain Member is a calcareous unit which forms the prominent cliffs and pinnacles along the canyon rim. USGS (2000) describes the Fossil Mountain Member as "light-gray, cliff-forming, fine- to medium-grained, thin- to medium- bedded [1-6 ft], fossiliferous, sandy, cherty limestone." Thickness of the Fossil Mountain Member has been reported ranging from 230 to 250 feet (USGS, 2000), or up to 250 to 300 feet (Hopkins and Thompson, 2003). The composition of this member transitions from predominantly limestone in the western Grand Canyon to increasingly siliceous to the east. Along the Hermit Trail approximately 10 miles southwest of the North Rim bore site, the unit is approximately 50% siliciclastic sandstone or sandy dolomite. This member is known to contain chert nodules throughout,

which typically occur as thin, continuous layers parallel to bedding about 2 inches thick (Huntoon, 1968; Hopkins and Thompson, 2003). The upper portion contains brecciated chert beds 4-10 feet thick where it forms the contact with the overlying Harrisburg Member (USGS, 2000).

The Fossil Mountain member forms an unconformable contact with the underlying Toroweap Formation, which can be attributed to erosive solutioning and channeling of the upper Toroweap. Localized infilling of these channels with sandy cherty limestone is typical (USGS, 2000).

3.2.2. Toroweap Formation

Generally, the Lower Permian Toroweap Formation includes pale orange to white fine-grained sandstone and siltstone separated by a limestone member, buff in color (Hargis & Montgomery, 1984). This Formation typically forms a vegetated slope and includes, in descending stratigraphic order, the Woods Ranch Member, Brady Canyon Member, and Seligman Member. Significant variability in lithology is characteristic of the Toroweap, with east-west lateral facies transitions occurring throughout the Grand Canyon and regionally (Turner, 2003). The total vertical thickness of the Toroweap Formation at the North Rim bore site is anticipated to be 310 feet based on available literature and evaluation of topographic slope breaks.

Solution tubes formed within limestone and gypsum beds, particularly along discontinuities within the Toroweap, result in a high secondary permeability which permits downward transmission of groundwater. Less permeable units within the Toroweap Formation cause lateral migration and formation of seeps typically on west-facing slopes (Huntoon, 1968).

3.2.2.1. Woods Ranch Member

The Woods Ranch Member consists of “gray and light-red, slope forming [gypsum-bearing] siltstone and silty sandstone, interbedded with white laminated gypsum and gray thin-bedded limestone” (USGS, 2000). Although USGS indicates gypsum beds up to 10 feet thick within the unit, a map of facies distribution provided in Turner (2003) suggests these evaporite beds are generally restricted to a region far northwest of the project area. Along the South Kaibab Trail, near the North Rim bore site, this member consists primarily of cross-bedded sandstone with wavy bedforms and brecciated carbonate beds at the base (Turner, 2003). This lithology has an overall lower occurrence of the carbonate and evaporite beds found further to the north and northwest.

USGS (2000) notes a unit thickness of about 65 ft in the southeastern region of the Grand Canyon. This unit is interpreted to have formed from transgressive inundation of the underlying Brady Canyon carbonates.

The gradational contact between the Woods Ranch member and the underlying Brady Canyon member consists of a dolomite bed with distinct desiccation cracks (Turner, 2003). The contact is defined topographically at the top of the limestone cliff of the Brady Canyon Member (USGS, 2000).

3.2.2.2. Brady Canyon Member

The Brady Canyon Member consists of “gray, cliff-forming, thin- to medium-bedded [1-5 ft] fine- to coarse-grained, fetid, fossiliferous limestone” with chert nodules comprising less than 8 percent of the

unit (USGS, 2000). The Brady Canyon member is a distinct cliff-forming carbonate unit which separates the Woods Ranch and Seligman Members. Towards the east near the North Rim bore site, dolomitic composition is typical. Deposition of this unit occurred in a shallow marine environment (Turner, 2003). A map of lateral facies transitions in the Brady Canyon member, from Turner (2003), indicates "sandy dolomite" as the primary lithology throughout the Kaibab Plateau where the site is located. Unit thickness ranges from about 40 ft in the eastern portion of the USGS (2000) quadrangle thickening to about 130 feet at the western edge.

The contact between the Brady Canyon Member and the underlying Seligman Member is marked by a gradational transition from the thick carbonate beds of the Brady Canyon member to the heterogeneous interbeds of the Seligman Member (Turner, 2003). The contact is defined topographically at the base of the limestone cliff of the Brady Canyon Member (USGS, 2000).

3.2.2.3. Seligman Member

The Seligman Member consists of "gray, light-purple, and yellowish-red, slope-forming, thin-bedded dolomite, sandstone, and gypsum" (USGS, 2000). The eastern portion of the USGS quadrangle in particular consists primarily of calcareous sandstone with thin limestone interbeds.

Despite the prevalence of evaporites northwest of the Grand Canyon within the Toroweap, a map of facies distribution provided in Turner (2003) shows a primary facies of "irregular-bedded sandstone" within the Seligman member on the Kaibab Plateau. This lithology has a lower occurrence of the carbonate and evaporite beds found further to the north and northwest. Thickness of the Seligman Member is about 30 feet in the eastern portion of the Grand Canyon increasing to about 55 feet to the west (USGS, 2000). Deposition of this unit occurred in a tidal flat environment (Turner, 2003).

The base of the Seligman member marks a conformable contact with underlying Coconino Sandstone and forms a sharp gradational boundary where the absence of cross bedding and/or presence of evaporites distinguishes the Seligman from the Coconino (USGS, 2000; Turner, 2003).

3.2.3. Coconino Sandstone

The Lower Permian Coconino Sandstone consists of "tan to white, cliff-forming, fine-grained, well-sorted, cross-bedded quartz sandstone" (USGS, 2000). Minor amounts of potassium feldspar are also present (Middleton et al., 2003). Hargis & Montgomery (1984) reported a thickness at the south rim of about 500 feet and indicated the unit is brittle and friable. The unit is well-lithified with siliceous cement and large-scale cross-stratified beds up to 66 feet thick. Uniform sorting and siliceous mineralogy provide evidence of an eolian depositional environment (Middleton et al., 2003).

The Coconino Sandstone forms a sharp planar disconformity with the underlying Hermit Formation (Blakey, 2003). This contact is marked by infilled desiccation cracks in the upper surface of the Hermit Formation (USGS, 2000).

Due to its lithology, the Coconino is a major aquifer throughout much of northern Arizona. Within the Grand Canyon region, however, it tends to bear water only where heavily fractured and where the underlying Hermit Formation forms an aquitard (Hargis & Montgomery, 1984).

3.2.4. Hermit Formation

The Lower Permian Hermit Formation, historically referred to as the Hermit Shale, consists of "red, slope-forming, fine-grained, thin-bedded siltstone and sandstone" (USGS, 2000). Blakey (2003) further specifies "reddish-brown siltstone, mudstone, and very fine-grained sandstone," with cyclically interbedded silty sandstone and sandy mudstone comprising the bulk of the unit. The Hermit Formation is about 260 feet thick in the eastern Grand Canyon and thickens to the west (USGS, 2000). Beds of silty sandstone form ledges approximately 3 feet on average and may contain calcareous nodules. Conversely, the interbedded sandy mudstone / platy siltstone is friable and forms slopes. A minor component of conglomerate is also present, with clasts largely consisting of calcareous concretions (Blakey, 2003).

Hargis & Montgomery (1984) reported a thickness at the south rim of about 250 feet. They also noted the difficulty of discerning the Hermit Formation from the underlying Supai Group in boring samples. This unit forms an unconformable contact with the underlying Esplanade Sandstone where it infills erosion channels up to 60 feet deep, but typically less than 10 feet (USGS, 2000).

The Hermit Formation behaves as an aquitard, which often results in perched groundwater conditions at the base of the overlying Coconino Sandstone (Hargis & Montgomery, 1984).

3.2.5. Supai Group

The Supai Group spans the Permian, Pennsylvanian, and Mississippian periods, and includes alternating slopes and benches of red siltstone and mudstone, light-brown fine-grained sandstone, and occasional limestone beds, some containing chert. Hargis & Montgomery (1984) reported a thickness at the south rim of about 930 feet. It includes, in descending stratigraphic order, the Esplanade Sandstone, Wescogame Formation, Manakacha Formation, and Watahomigi Formation.

3.2.5.1. Esplanade Sandstone

The Lower Permian Esplanade Sandstone consists of an upper "light-red and pinkish-gray, cliff-forming, fine- to medium-grained, medium-bedded [3-10 ft], well-sorted calcareous sandstone" with a lower slope-forming unit consisting of dark-red thinly-bedded siltstone (USGS, 2000). The Esplanade Sandstone is emplaced between the steep ledges and slopes of the underlying Wescogame and the more continuous slopes of the Hermit Formation above. Of all the Supai group units, the Esplanade contains the highest fraction of sandstone. In the eastern Grand Canyon, the Esplanade ranges from 200 to 250 feet thick, although measured sections along the Kaibab Trail are about 300 ft (Blakey, 2003). The lithology comprising the Esplanade is dominated by cross-stratified sandstone exhibiting evidence of eolian deposition. Thinly bedded limestone may also occur in the lower slope-forming portion (USGS, 2000). Clean surfaces appear pale grayish orange to reddish orange, although outcrops are typically stained red from the overlying mudstone (Blakey, 2003).

The contact with the underlying Wescogame Formation is unconformable and demarcated by channels infilled with limestone conglomerate which average about 35 feet deep (USGS, 2000).

3.2.5.2. Wescogame Formation

USGS (2000) describes the Upper Pennsylvanian Wescogame Formation as “light-red, pale-yellow, and light-gray upper slope unit and lower cliff unit.” The upper slope-forming subunit consists primarily of fine-grained siltstone and mudstone with heterogeneous interbeds of calcareous sandstone and conglomerate. The lower subunit consists primarily of cross-bedded sandstone and calcareous sandstone, which forms cliffs up to 40 ft thick (USGS, 2000).

Total thickness of the Wescogame ranges from about 100 to 200 feet in the Grand Canyon and is about 130 feet thick along the Kaibab Trail (Blakey, 2003). The lithology of the Wescogame is predominately sandstone throughout much of the central Grand Canyon, with mudstone becoming more prevalent to the east.

The boundary between the Wescogame and Manakacha can be poorly defined, although basal conglomerate in the Wescogame often demarks the contact (Blakey, 2003). Topographically, the transition from the Wescogame can be determined by locating the end of the slope at the top of the Manakacha. The contact is unconformable and often consists of erosion channels infilled with limestone and chert conglomerate (USGS, 2000).

3.2.5.3. Manakacha Formation

USGS (2000) describes the Middle Pennsylvanian Manakacha Formation as “light-red, white, and gray upper slope unit and lower cliff unit of sandstone, calcareous sandstone, dark-red siltstone, and gray limestone.” The slope-forming unit consists of primarily siltstone and mudstone with some interbeds of limestone and sandstone. The cliff-forming unit consists of cross-bedded calcareous sandstone, dolomite, and sandy limestone. The upper unit is approximately 100 feet thick, and the lower cliff-forming unit is about 60 feet thick in the eastern Grand Canyon (USGS, 2000).

Within the Manakacha, three distinct types of sandstone are recognized by Blakey (2003). The first consists of variably thick cross-stratified bedding sets with indications of eolian deposition. Another type of sandstone consists of, “horizontally laminated to very low-angle, cross-stratified, fine-grained, calcareous and silty units up to 20 feet thick.” The third type of sandstone is grayish-orange to bright reddish-orange and consists of very fine-grained sands. The uppermost portions of the unit may have undergone significant chemical alteration resulting in the replacement of the original rock fabric via crystallization of calcite and dolomite (Blakey, 2003). The unconformable contact between the Manakacha and Watahomigi is demarcated by conglomerate containing fragments of the Watahomigi with the bedding surface exhibiting scour and channeling (USGS, 2000).

3.2.5.4. Watahomigi Formation

The Lower Pennsylvanian to Upper Mississippian Watahomigi Formation consists of “gray and purplish-red, slope-forming limestone, siltstone, mudstone, and conglomerate” within a 70-feet thick upper unit consisting of thin interbeds of cherty limestone, siltstone, and mudstone (USGS, 2000). The lower unit predominately forms slopes containing mudstone and siltstone with thin interbeds of limestone and red chert veins (USGS, 2000). Carbonates in this unit within the eastern Grand Canyon tend to be very fine-grained, with some larger fragments of fossils and pellets (Blakey, 2003). Mudstones within this

Formation tend to lack structure and have occasional interbeds of bright-orange calcareous sandstone. Blakey (2003) indicated two units of red slope-forming mudstone, separated by a ledge of carbonates. The Watahomigi averages about 100 feet thick in the eastern Grand Canyon (USGS, 2000; Blakey, 2003).

The Watahomigi forms an abrupt, unconformable contact with the underlying Redwall Limestone demarcated by interbedded conglomerates which infill erosion channels (USGS, 2000; Blakey, 2003).

3.2.6. Redwall Limestone

The Mississippian Redwall Limestone is light-gray and grayish-blue cliff-forming, crystalline limestone, but is typically stained red surficially due to wash from the overlying Supai Formation (Beus, 2003a). Deposition of the Redwall Limestone took place in a shallow marine environment and records two major cycles of transgression and regression. Bedding planes containing lenses and nodules of chert are common. Hargis & Montgomery (1984) reported an overall thickness of about 400 feet at the south rim. They also report cavernous conditions commonly occurring in the upper portion of the Formation. The karstic character of the Redwall allows for significant groundwater transmission and storage. The Redwall includes, in descending stratigraphic order, the Horseshoe Mesa, Mooney Falls, Thunder Springs, and Whitmore Wash Members.

The Redwall Limestone contains small solution tubes that are infilled with redbed deposits in the upper portion of the Formation. Due to its solubility, it may develop small voids along discontinuities which permit moderate transmission of groundwater and form small seeps within the formation (Huntoon, 1968).

3.2.6.1. Horseshoe Mesa Member

The Horseshoe Mesa Member consists of “light-olive-gray, ledge- and cliff-forming, thin-bedded, fine-grained limestone” USGS (2000). This is the thinnest member of the Redwall, averaging about 45 feet thick in the eastern Grand Canyon. The lower portion of the unit contains lenses of chert. This unit forms a receding sequence of weak ledges which contrast with the underlying massive vertical cliffs (Beus, 2003a).

The Horseshoe Mesa Member forms a conformable contact with the underlying Mooney Falls Member, which can be difficult to locate due to the similarity in lithology between the two members. A transition from a finely textured, thin-bedded limestone to a medium-coarse-grained, massive limestone demarks the contact (Beus, 2003a).

3.2.6.2. Mooney Falls Member

The Mooney Falls Member consists of “light-gray, cliff-forming, fine- to coarse-grained, thick-bedded to very thick-bedded [4-20 feet], fossiliferous limestone” (USGS, 2000). The Mooney Falls Member is the thickest member of the Redwall and is about 200 feet thick in the eastern Grand Canyon. The pure limestone that forms the bulk of this unit includes invertebrate fossils, oolites, pellets and skeletal fragments. The upper portion of the Mooney Falls contains isolated zones of chert in thin beds or lenses (Beus, 2003a).

Minor structural deformation and erosion of the underlying Thunder Springs Member forms a low-angle unconformity at the base of the Mooney Falls Member.

3.2.6.3. Thunder Springs Member

The Thunder Springs Member consists of "[approximately half] gray, cliff-forming, fossiliferous, thin-bedded limestone, and [the other half] brownish-gray, cliff-forming, thin-bedded [1-5 inches], finely crystalline dolomite and fine- to coarse-grained limestone interbedded with white beds of chert lenses" (USGS, 2000). Distinct alternating bands of gray limestone or dolomite, and chert distinguish the unit. It is about 100 feet thick in the eastern Grand Canyon. In the eastern Grand Canyon, carbonate composition is typically dolomitic (Beus, 2003a).

The Thunder Springs Member forms a conformable contact with the underlying Whitmore Wash Member. Thunder Springs can be easily differentiated from the pure carbonates of the Whitmore Wash by the distinct light-dark banding of chert and dolomite (Beus, 2003a).

3.2.6.4. Whitmore Wash Member

The Whitmore Wash Member consists of "yellowish-gray and brownish-gray, cliff-forming, thick-bedded, fine-grained dolomite" (USGS, 2000). It is about 100 feet thick in the eastern Grand Canyon. This member is formed of thick beds (2 to 4 feet typically) of mostly pure carbonate, with a minor component of gypsum and iron oxides (Beus, 2003a).

The contact of the Whitmore Wash Member with the underlying Temple Butte Formation constitutes a significant unconformity at the base of the Redwall Limestone. The contact is demarcated by a basal conglomerate containing clasts of angular dolomite or limestone from the Temple Butte Formation, typically forming a horizontal surface (Beus, 2003a). The contact can be difficult to recognize where it separates two dolomitic units. The two units can be distinguished topographically by the transition from the steep resistant cliff of the Redwall Limestone, overlying a series of receding ledges typical of the Temple Butte.

3.2.7. Temple Butte Formation

The Devonian Temple Butte Formation consists of "purple, reddish-purple, dark-gray, and light-gray, ledge-forming dolomite, sandy dolomite, sandstone, mudstone, and limestone" (USGS, 2000). Hargis & Montgomery (1984) reported a unit thickness of about 75 feet at the south rim. This unit thickness is consistent with an interpolated thickness between selected stratigraphic sections of the Temple Butte Formation provided by Beus (2003b).

This unit tends to occur as thin, discontinuous lenses in the eastern Grand Canyon, which thicken and become more continuous towards the west. The Temple Butte typically infills eroded channels within the Upper Muav Formation, forming lenses which can be up to 400 feet wide, but typically less than 100 feet thick (USGS, 2000).

The Temple Butte Formation can contain small solution channels which daylight as low-discharge springs in areas. The location of solution channels is controlled by the location of discontinuities within the Formation (Huntoon, 1968).

3.2.8. Tonto Group

The Cambrian Tonto Group in the Grand Canyon is demarcated by a large-scale prominent bench known as the Tonto Platform, which follows the top of the Tapeats Sandstone. Generally, the Tonto Group comprises a system of sandstone, mudstone, and limestone which forms a series of small cliffs, separated by thicker intervals of the slope-forming Bright Angel Formation (Middleton and Elliot, 2003).

Sedimentation of the Tonto Group, and the Tapeats Sandstone in particular, have been heavily influenced by the relief of the pre-deposition topography along the Great Unconformity. The upper horizon of the Precambrian strata includes evidence of significant weathering prior to deposition of the Tonto Group. The thickness of this weathering horizon, where present, ranges from less than 10 feet up to 50 feet thick (Middleton and Elliot, 2003).

The Cambrian rocks of the Tonto group constitute a transgressive sequence, representing the transition from clastic shoreline deposits to marine carbonates as the shoreline migrated eastward. However, the overall eastward transgression was interspersed with intermediate periods of regression, which resulted in a relatively complex interchange between the various lithologies. The Tonto group generally thickens to the west, likely attributable to seaward subsidence during deposition (Middleton and Elliot, 2003).

Units in the Tonto group include (in descending stratigraphic order) the Muav Formation, the Bright Angel Formation, and the Tapeats Sandstone.

3.2.8.1. Muav Formation

The Middle Cambrian Muav Formation consists of “dark-gray, light-gray, brown, and orange-red, cliff-forming limestone, dolomite, and calcareous mudstone” (USGS, 2000). Texturally, the Muav is typically mottled packstone with interbeds of flat-pebble conglomerate. Flat-pebble conglomerates in particular are abundant east of the Bass Trail and consist of “disc-like clasts of micrite and, occasionally, silt-size quartz and glauconite” (Middleton and Elliot, 2003). Within the massive carbonate beds are occasional thin beds of micaceous shale, siltstone, and fine-grained sandstone. The fraction of siliciclastics within the Muav increases toward the east with a corresponding decrease in carbonates. Hargis & Montgomery (1984) reported a unit thickness of about 350 feet at the south rim. They also indicated the presence of karstic features in the lower 250 feet. The Muav Formation was primarily deposited in a subtidal environment (Middleton and Elliot, 2003).

A complex intertonguing gradational contact forms the conformable boundary between the Muav Formation the underlying Bright Angel Formation. This intertonguing contact marks a transition from the shallow-water facies of the Bright Angel Formation to the sub-tidal environment associated with the Muav (Middleton and Elliot, 2003).

The solubility of the Muav Formation results in solution features up to 25 feet in diameter, which permits significant storage and transmission of groundwater. The presence of solution cavities is largely dependent on the location of discontinuities and can vary significantly. Downward transmission of groundwater is limited by the underlying Bright Angel Formation causing lateral migration within the Muav. The Muav contains large karstic springs (such as Roaring Springs) with high peak discharges exceeding 150 cubic feet per second (cfs) (Huntoon, 1968).

3.2.8.2. Bright Angel Formation

The Middle Cambrian Bright Angel Formation consists of “green and purple-red, slope-forming siltstone and shale, and interbeds of red-brown to brown sandstone of Tapeats Sandstone lithology” (USGS, 2000). Although the Bright Angel Formation is lithologically complex, the primary lithology is fossiliferous shale composed largely of clay minerals and stained green from the presence of glauconite. Conglomerates and coarse-grained sandstones are interspersed throughout the unit containing various amounts of quartz, potassium feldspar, and sedimentary rock fragments. A unit thickness of approximately 325 feet has been measured near Bright Angel Creek (Middleton and Elliot, 2003). The depositional environment of the Bright Angel Formation is primarily subtidal, with evidence of occasional disturbance from storm-induced currents.

The conformable contact with the underlying Tapeats Sandstone is gradational and can be difficult to discern. Topographic features are likely the best indicator of the transition, with the contact roughly located at the transition from the slope-forming Bright Angel Formation to the cliff-forming Tapeats Sandstone (Middleton and Elliot, 2003).

The Bright Angel Formation is hydrologically impervious and limits downward migration of groundwater. This creates an effective aquiclude which significantly influences groundwater accumulation in the overlying Muav (Huntoon, 1968).

3.2.8.3. Tapeats Sandstone

The Middle and Lower Cambrian Tapeats Sandstone consists of “brown and red-brown, cliff-forming sandstone and conglomerate” (USGS, 2000). The Tapeats Sandstone will not be encountered in the bores and therefore is not included in Figure 5.1. The sandstone composition is rich in quartz and medium- to coarse-grained feldspar. Pebble-size conglomerates tend to occur near the base of the formation and are also rich in quartz. Typical outcrops of the Tapeats occur as cliff-forming beds averaging less than 3 feet thick. Thinner interbeds of sandstone and mudstone are emplaced above the main cliff, indicating a facies transition to the overlying Bright Angel Formation. The thickness of the Tapeats is variable throughout the Grand Canyon, with an estimated maximum thickness of 390 feet along the Bass Trail. Typical thicknesses are within the range of 100 to 325 feet and are heavily influenced by the underlying Precambrian paleo-topography. The Tapeats was deposited primarily in a tidal flat environment, with evidence of shoreline and fluvial deposition (Middleton and Elliot, 2003).

The Great Unconformity defines the contact between the Tapeats Sandstone and the underlying Proterozoic Rocks, marking a significant gap in the geologic record below this unit.

The Tapeats Sandstone is hydrologically insignificant due to the overlying Bright Angel Formation aquiclude and its relatively lower permeability (Huntoon, 1968).

4. Anticipated Geologic Conditions and Ground Behavior

4.1. General

The bedrock characteristics and ground conditions described in this section of the GISR constitute the interpreted geological conditions anticipated to be encountered during directional drilling. Geologic interpretations described herein include characteristics of each formation and location of geologic features relevant to directional drilling, bore stability in each geologic formation, strength and abrasivity of rock types anticipated to be encountered, and hydrogeologic features and groundwater conditions anticipated to be encountered during directional drilling.

The interpreted geologic profile for the bore path is shown on Figure 5-1. The interpretations included in this report were developed using data obtained during field investigations, the geophysical investigation, and information obtained from the literature review as described herein. Interpretations for bore stability are derived from the Tunnelman's classification system for rock which is included in Appendix A.

4.2. Geologic Formation Characteristics

This section describes the defining characteristics and relevant geologic features for each formation and member in the geologic sequence anticipated to be encountered during directional drilling. Interpreted percentages of rock types present within each formation or member are provided. Additional interpretations regarding the potential for solution cavities are provided for karstic formations. Thorough descriptions of the lithologies within each formation are included in Section 3.2. Estimations of strength and abrasivity of individual rock types are provided in Section 4.3.

4.2.1. Kaibab Formation

The Harrisburg Member of the Kaibab Formation is not anticipated to be encountered during drilling. Only the Fossil Mountain Member of the Kaibab Formation is anticipated to be encountered during drilling. Interpreted characteristics of the Fossil Mountain Member are provided below.

4.2.1.1. Fossil Mountain Member

The Fossil Mountain Member exhibits cliff-forming surface weathering behavior. The Member is heavily fractured and subject to solutioning resulting in karst features including voids and enlarged fractures. However, based on the results of the 2024 ERT Survey within the Fossil Mountain Member, karstic solutioning and fractures larger than 40 feet in length are not anticipated to be encountered. Ground behavior in accordance with the Tunnelman's Ground Classification System for the Member is Blocky and Seamy Rock. Table 4-1 provides estimated percentages of rock types anticipated to be encountered within the Fossil Mountain Member.

Table 4-1. Estimated Rock Type Percent by Volume of Fossil Mountain Member

Rock Type	Percent by Volume (%)
Limestone/Dolomite	50
Sandstone	30
Chert	20

4.2.2. Toroweap Formation

The Toroweap Formation includes three members including the Woods Ranch Member, Brady Canyon Member, and Seligman Member. Contacts between each member are gradational. The following sections provide interpreted characteristics of each member of the Toroweap Formation.

4.2.2.1. Woods Ranch Member

The Woods Ranch Member may contain gypsum beds less than 10 feet thick, which are significantly softer than the surrounding rock types. The overall surface weathering behavior of the Member is slope-forming. Though solutioning within the gypsum and limestone beds is possible, no large voids are anticipated to be encountered within the Woods Ranch Member during drilling. Ground behavior in accordance with the Tunnelman's Ground Classification System for the Member is Stratified Rock. Table 4-2 provides estimated percentages of rock types anticipated to be encountered within the Woods Ranch Member.

Table 4-2. Estimated Rock Type Percent by Volume of Woods Ranch Member

Rock Type	Percent by Volume (%)
Siltstone	40
Sandstone	30
Gypsum	20
Limestone	10

4.2.2.2. Brady Canyon Member

The Brady Canyon Member exhibits a cliff-forming surface weathering behavior. The member is subject to solutioning resulting in karst features such as voids and enlarged fractures. Although limited solution cavities may be present, based on the results of the 2024-2025 Seismic Reflection Survey, karstic voids larger than 40 feet in length are not anticipated to be encountered within the Brady Canyon Member. Ground behavior in accordance with the Tunnelman's Ground Classification System for the Member is Blocky and Seamy Rock. Table 4-3 provides estimated percentages of rock types anticipated to be encountered within the Brady Canyon Member.

Table 4-3. Estimated Rock Type Percent by Volume of Brady Canyon Member

Rock Type	Percent by Volume (%)
Limestone	90
Chert	10

4.2.2.3. Seligman Member

The Seligman Member exhibits a slope-forming surface weathering behavior. Karst features are minor in the Seligman member and will not influence drilling. The base of the Seligman Member forms a gradational contact with the underlying Coconino Sandstone. Ground behavior in accordance with the Tunnelman's Ground Classification System for the Member is Stratified Rock. Table 4-4 provides estimated percentages of rock types anticipated to be encountered within the Seligman Member.

Table 4-4. Estimated Rock Type Percent by Volume of Seligman Member

Rock Type	Percent by Volume (%)
Sandstone	70
Limestone	25
Gypsum	5

4.2.3. Coconino Sandstone

The Coconino Sandstone is a sandstone unit with cliff-forming surface weathering behavior. The Coconino Sandstone is highly fractured including open fractures where the unit is well cemented. The Coconino Sandstone forms a sharp contact with the underlying Hermit Formation. Ground behavior in accordance with the Tunnelman's Ground Classification System for the Formation is Moderately Jointed Rock. Table 4-5 provides estimated percentages of rock types anticipated to be encountered within the Coconino Sandstone.

Table 4-5. Estimated Rock Type Percent by Volume of Coconino Sandstone

Rock Type	Percent by Volume (%)
Sandstone	100

4.2.4. Hermit Formation

The Hermit Formation exhibits alternating surface weathering behavior including ledge-forming behavior in sandstone beds and slope forming behavior in the mudstone beds. The overall surface expression of the Formation is slope-forming. The contact with the underlying Wescogame Formation is unconformable and difficult to discern from surface expression. Ground behavior in accordance with the Tunnelman's Ground Classification System for the Formation is Moderately Jointed Rock. Table 4-6 provides estimated percentages of rock types anticipated to be encountered within the Hermit Formation.

Table 4-6. Estimated Rock Type Percent by Volume of Hermit Formation

Rock Type	Percent by Volume (%)
Sandstone	35
Shale	35
Siltstone	25
Conglomerate	5

4.2.5. Supai Group

4.2.5.1. Esplanade Sandstone

The Esplanade Sandstone consists of an upper calcareous sandstone unit and a lower siltstone unit. The upper and lower units exhibit cliff-forming and slope forming surface weathering behavior, respectively. Ground behavior in accordance with the Tunnelman's Ground Classification System for the upper and lower units are Intact Rock and Stratified Rock, respectively. Table 4-7 provides estimated percentages of rock types anticipated to be encountered within the Esplanade Sandstone.

Table 4-7. Estimated Rock Type Percent by Volume of Esplanade Sandstone

Rock Type	Percent by Volume (%)	
	Upper Unit	Lower Unit
Sandstone	90	10
Siltstone	10	85
Conglomerate	0	5

4.2.5.2. Wescogame Formation

The Wescogame Formation consists of an upper siltstone, mudstone, and sandstone unit and lower sandstone unit. The upper and lower units exhibit slope-forming and cliff-forming surface weathering behavior, respectively. Ground behavior in accordance with the Tunnelman's Ground Classification System for the upper and lower units are Stratified Rock and Moderately Jointed Rock, respectively. Table 4-8 provides estimated percentages of rock types anticipated to be encountered within the Wescogame Formation.

Table 4-8. Estimated Rock Type Percent by Volume of Wescogame Formation

Rock Type	Percent by Volume (%)	
	Upper Unit	Lower Unit
Mudstone	40	0
Siltstone	40	5
Sandstone	15	90
Conglomerate	5	5

4.2.5.3. Manakacha Formation

The Manakacha Formation consists of an upper siltstone, mudstone, and sandstone unit and lower sandstone and limestone unit. The upper and lower units exhibit slope-forming and cliff-forming surface weathering behavior, respectively. Ground behavior in accordance with the Tunnelman's Ground Classification System for the upper and lower units are Stratified Rock and Moderately Jointed Rock, respectively. Table 4-9 provides estimated percentages of rock types anticipated to be encountered within the Manakacha Formation.

Table 4-9. Estimated Rock Type Percent by Volume of Manakacha Formation

Rock Type	Percent by Volume (%)	
	Upper Unit	Lower Unit
Mudstone	35	0
Siltstone	35	5
Sandstone	15	45
Limestone	10	45
Conglomerate	5	5

4.2.5.4. Watahomigi Formation

The Watahomigi Formation consists of an upper limestone with interbedded siltstone and mudstone unit and lower mudstone and siltstone with interbedded limestone unit. The upper and lower units exhibit slope- and ledge-forming and slope-forming surface weathering behavior, respectively. Ground behavior in accordance with the Tunnelman's Ground Classification System for the upper and lower units are Blocky and Seamy Rock and Stratified Rock, respectively. Table 4-10 provides estimated percentages of rock types anticipated to be encountered within the Watahomigi Formation.

Table 4-10. Estimated Rock Type Percent by Volume of Watahomigi Formation

Rock Type	Percent by Volume (%)	
	Upper Unit	Lower Unit
Mudstone	20	30
Siltstone	20	30
Limestone	40	25
Chert	15	10
Conglomerate	5	5

4.2.6. Redwall Limestone

The Redwall Limestone includes four members; the Horseshoe Mesa Member, the Mooney Falls Member, the Thunder Springs Member, and the Whitmore Wash Member. Contacts between members are primarily defined by unconformities and minor changes in lithology. Due to the similarities in lithology, solutioning occurs throughout all four members of the Redwall Limestone resulting in karst features such as voids and enlarged fractures that occur most commonly in the upper third of the Formation. For this reason, interpretations of karstic features are undifferentiated between the four members of the Redwall limestone. Karstic voids or small solution cavities may be present within the Redwall Limestone, however, based on the results of seismic reflection imaging summarized in the North Rim Geophysical Studies Report (GEI, Collier, GKS; 2025), karstic features larger than 40 feet in length which would adversely impact drilling are not anticipated to be encountered within the Redwall Limestone along the bore paths. The following sections provide interpreted characteristics of each member of the Redwall Limestone.

4.2.6.1. Horseshoe Mesa Member

The Horseshoe Mesa Member exhibits cliff-forming surface weathering behavior. The contact with the underlying Mooney Falls Member is conformable and difficult to discern. Ground behavior in accordance with the Tunnelman's Ground Classification System for the Formation is Moderately Jointed Rock. Table 4-11 provides the estimated percentages of rock types anticipated to be encountered in the Horseshoe Mesa Member.

Table 4-11. Estimated Rock Type Percent by Volume of Horseshoe Mesa Member

Rock Type	Percent by Volume (%)
Limestone	90
Chert	10

4.2.6.2. Mooney Falls Member

The Mooney Falls Member exhibits cliff-forming surface weathering behavior. The contact with the underlying Thunder Springs Member forms a low-angle angular unconformity. Ground behavior in accordance with the Tunnelman's Ground Classification System for the Formation is Moderately Jointed Rock. Table 4-12 provides the estimated percentages of rock types anticipated to be encountered in the Mooney Falls Member.

Table 4-12. Estimated Rock Type Percent by Volume of Mooney Falls Member

Rock Type	Percent by Volume (%)
Limestone	90
Chert	10

4.2.6.3. Thunder Springs Member

The Thunder Springs Member exhibits cliff forming surface weathering behavior. The contact with the underlying Whitmore Wash Member is planar and conformable. Ground behavior in accordance with the Tunnelman's Ground Classification System for the Formation is Moderately Jointed Rock. Table 4-13 provides the estimated percentages of rock types anticipated to be encountered in the Thunder Springs Member.

Table 4-13. Estimated Rock Type Percent by Volume of Thunder Springs Member

Rock Type	Percent by Volume (%)
Limestone/Dolomite	80
Chert	20

4.2.6.4. Whitmore Wash Member

The Whitmore Wash Member exhibits a cliff-forming surface weathering behavior. The lower contact of the Member with the underlying Temple Butte Formation is horizontal and unconformable. Ground behavior in accordance with the Tunnelman's Ground Classification System for the Formation is

Moderately Jointed Rock. Table 4-14 provides estimated percentages of rock types anticipated to be encountered in the Whitmore Wash Member.

Table 4-14. Estimated Rock Type Percent by Volume of Whitmore Wash Member

Rock Type	Percent by Volume (%)
Limestone/Dolomite	90
Gypsum	5
Conglomerate	5

4.2.7. Temple Butte Formation

The Temple Butte Formation may contain small solution channels due to solutioning of limestone. However, large karstic voids which would adversely impact drilling are not anticipated to be encountered within the Temple Butte Formation. The Formation exhibits ledge-forming surface weathering behavior. Ground behavior in accordance with the Tunnelman's Ground Classification System for the Formation is Moderately Jointed Rock. Table 4-15 provides estimated percentages of rock types anticipated to be encountered in the Temple Butte Formation.

Table 4-15. Estimated Rock Type Percent by Volume of Temple Butte Formation

Rock Type	Percent by Volume (%)
Limestone	55
Sandstone	40
Conglomerate	5

4.2.8. Tonto Group

4.2.8.1. Muav Formation

The Muav Formation is subject to solutioning forming karstic features such as voids and enlarged fractures. Karstic voids or small solution cavities may be present within the Muav Formation, however, based on the results of seismic reflection imaging summarized in the North Rim Geophysical Studies Report (GEI, Collier, GKS; 2025), karstic features larger than 40 feet in length which would adversely impact drilling are not anticipated to be encountered within the Muav Formation along the bore paths. The Formation exhibits cliff-forming surface weathering behavior. The underlying Bright Angel Formation forms a gradational contact that intertongues with the Muav Formation. Ground behavior in accordance with the Tunnelman's Ground Classification System for the Formation is Blocky and Seamy Rock. Table 4-16 provides estimated percentages of rock types anticipated to be encountered in the Muav Formation.

Table 4-16. Estimated Rock Type Percent by Volume of Muav Formation

Rock Type	Percent by Volume (%)
Limestone/Dolomite	60
Mudstone	20
Conglomerate	10

Rock Type	Percent by Volume (%)
Siltstone	5
Sandstone	5

4.2.8.2. Bright Angel Formation

The Bright Angel Formation exhibits slope-forming surface weathering behavior. Sandstone interbeds of Tapeats Sandstone Lithology are present in the lower portion of the formation. Ground behavior in accordance with the Tunnelman's Ground Classification System for the Formation is Stratified Rock. Table 4-17 provides estimated percentages of rock types anticipated to be encountered in the Bright Angel Formation.

Table 4-17. Estimated Rock Type Percent by Volume of Bright Angel Formation

Rock Type	Percent by Volume (%)
Shale	40
Siltstone	30
Sandstone	25
Conglomerate	5

4.3. Rock Properties

Rock types encountered along the bore path are grouped into seven rock types including mudstone/shale, gypsum, siltstone, limestone/dolomite, conglomerate, sandstone, and chert. Rock types are grouped by their primary lithological components. Mudstone/shale and limestone/dolomite have distinct geologic classifications based on rock texture, chemistry, and depositional environment, but are grouped due to their similar lithology and respective similarities in engineering properties. Though minor differences in composition occur between geologic formations, specific rock types are assumed to exhibit the same behavior with respect to strength and abrasivity. Estimations for compressive strength and abrasivity were informed by compressive strength testing performed for the North Rim Investigation, Schmidt hammer testing on rock samples collected during the Inner Canyon investigation, review of available literature, and experience in similar lithologies. Specific literature used to determine compressive strength values included Hunt (1984) and Farmer and Attwell (1976). Abrasivity values are based on index ranges provided in ASTM D7625 Standard Test Method for Laboratory Determination of Abrasiveness of Rock Using the CERCHAR Abrasiveness Index Method. Table 4-18 provides estimated strength and abrasivity parameters for each rock type anticipated to be encountered during directional drilling.

Table 4-18. Estimated Strength and Abrasivity Parameters by Rock Type

Rock Type	Estimated Compressive Strength (psi)	Estimated Abrasivity
Mudstone/Shale	100 – 5,000	Very Low
Gypsum	100 – 5,000	Very Low to Low
Siltstone	100 – 5,000	Low to Medium
Limestone/Dolomite	5,000 – 15,000	Medium to High

Rock Type	Estimated Compressive Strength (psi)	Estimated Abrasivity
Conglomerate	5,000 – 16,000	Medium to Very High
Sandstone	4,000 – 16,000	High to Very High
Chert	15,000 – 35,000	Very High to Extremely High

4.4. Anticipated Hydrogeologic Conditions

4.4.1. Groundwater

Two aquifers are known to exist in Northern Arizona including the upper C-aquifer and the lower R-aquifer. The C-aquifer includes the Kaibab Formation, Coconino Sandstone, and Supai Group with the lower strata of the Supai Group forming an aquitard with the formations below. The R-aquifer includes the Redwall Limestone, Temple Butte Formation, and Muav Formation with the Bright Angel Formation forming a lower aquitard. Flow occurs between these two aquifers through fractures and faults in the rock mass. A CSAMT survey performed by Zonge unrelated to this contract approximately 2 miles northwest of the bore location indicated the presence of a strong R-aquifer with flow between the C-aquifer and R-aquifer occurring through discontinuities in the lower strata of the Supai Group (Zonge, 2015). The approximate groundwater tables anticipated to be encountered within the C- and R-aquifers are shown in Figure 5-1. Results of the seismic reflection imaging summarized in the 2025 North Rim Geophysical Studies Report (GEI, Collier, GKS; 2025) confirmed the lack of a major confined water bearing unit which would exhibit high porewater pressures. Due to the nature of groundwater within the formations, water is likely isolated to discontinuous fractures or dissolution cavities. Therefore, groundwater within these formations will not develop a large hydrostatic head and resulting pore pressures will be nominal.

4.4.2. Hydraulic Conductivity and Cement Take

The flow of water within the rock mass is a function of the primary hydraulic conductivity and secondary hydraulic conductivity of the formations. Primary hydraulic conductivity is a function of grain size, density, and matrix composition, and secondary hydraulic conductivity is a function of the orientation, quantity, and size of water-transmitting discontinuities. Hydraulic conductivity affects fluid losses and pore pressures during drilling and cement take during cementing. Table 4-19 provides estimated hydraulic properties including primary hydraulic conductivity and porosity of each formation anticipated to be encountered during drilling.

The estimated hydraulic conductivity values provided do not consider secondary permeability due to fracturing in the rock mass nor the size and quantity of solution cavities within karstic formations. Groundwater flow in the vicinity of the bore path is anticipated to occur primarily along vertical or subvertical joints and horizontal bedding planes within the rock mass. Increased influx of groundwater during drilling would be primarily expected where the bore path intersects a water-transmitting discontinuity. Groundwater flow direction is expected to generally follow bedding planes which dip gradually to the southwest. Seeps observed by GEI and others during field reconnaissance in Transept Canyon immediately southwest of the bore path had minimal groundwater discharge with no water accumulation at the base of the canyon walls. This is indicative of low pore pressures and minimal

groundwater transmission within the rock mass at this location, and significant groundwater influx due to fracture flow is not anticipated to be encountered. The contractor should use the information contained in Sections 4.2 and 5.5 to inform an estimate of additional cement quantities that will be required due to fracturing in the rock mass and the potential for intercepting solution cavities in karstic formations.

Table 4-19. Estimated Hydraulic Properties

Formation	Member	Primary Hydraulic Conductivity (m/s)[1]	Porosity (%)[2]
Kaibab Formation	Fossil Mountain	10^{-6} – 10^{-2}	5 - 20
Toroweap Formation	Woods Ranch	10^{-11} – 10^{-7}	10 – 30
	Brady Canyon	10^{-9} – 10^{-6}	5 – 25
	Seligman	10^{-10} – 10^{-6}	5 - 25
Coconino Sandstone		10^{-10} – 10^{-6}	5 – 25
Hermit Formation		10^{-12} – 10^{-6}	10 – 30
Supai Group - Esplanade Sandstone		10^{-11} – 10^{-6}	5 – 25
Supai Group - Wescogame Formation		10^{-11} – 10^{-6}	5 – 25
Supai Group - Manakacha Formation		10^{-11} – 10^{-6}	5 – 25
Supai Group - Watahomigi Formation		10^{-11} – 10^{-7}	10 – 30
Redwall Limestone	Horseshoe Mesa	10^{-6} – 10^{-2}	5 – 20
	Mooney Falls	10^{-6} – 10^{-2}	5 – 20
	Thunder Springs	10^{-6} – 10^{-2}	5 – 20
	Whitmore Wash	10^{-6} – 10^{-2}	5 – 20
Temple Butte Formation		10^{-9} – 10^{-3}	5 – 20
Tonto Group - Muav Formation		10^{-6} – 10^{-2}	5 – 20
Tonto Group - Bright Angel Formation		10^{-11} – 10^{-6}	10 - 30

Notes:
1. From Freeze and Cherry, 1979
2. From Attwell and Farmer, 1976

5. Construction Considerations

5.1. General

This section discusses construction considerations and potential issues the Contractor may encounter with respect to the anticipated geological conditions discussed in the previous section. The section summarizes anticipated drilling conditions, casing pipe design, drill bit considerations, and cementing. Anticipated drilling conditions included herein and shown in Figure 5-1 were assessed based on literature review and in-depth analysis of the surface weathering characteristics of each formation using USGS LiDAR data. Construction considerations provided below as well as additional drilling considerations are discussed in greater detail in the Bore Drilling Plan (GKS, 2026).

The contractor is responsible for all temporary aspects of construction and determining the means and methods required to complete the work within the limitations provided in the Project Specifications. The Contractor is responsible for providing comprehensive submittals to the Engineer in accordance with the Project Specifications.

5.2. Anticipated Drilling Conditions

The bore is anticipated to encounter 10 feet of soil at the bore entry location that will be removed during spudding. The Contractor is then anticipated to encounter the geologic formations as described in Section 4.2 of this Report. Table 5-1 provides the anticipated drilling conditions along the bore path.

Table 5-1. Anticipated Drilling Conditions

Formation	Member	Vertical Thickness (ft)	Contact Elevation (ft)	True Vertical Contact Depth (ft)[1]	Measured Contact Depth (ft)[2]
Kaibab Formation	Fossil Mountain	247	8,237	10	10
Toroweap Formation	Woods Ranch	215	7,990	248	248
	Brady Canyon	50	7,775	463	463
	Seligman	45	7,725	513	513
Coconino Sandstone		365	7,680	558	558
Hermit Formation		365	7,315	923	923
Supai Group - Esplanade Sandstone		375	6,950	1,288	1,288
Supai Group - Wescogame Formation		110	6,575	1,663	1,665
Supai Group - Manakacha Formation		280	6,465	1,773	1,781
Supai Group - Watahomigi Formation		125	6,185	2,053	2,105
Redwall Limestone	Horseshoe Mesa	70	6,060	2,178	2,283
	Mooney Falls	245	5,990	2,248	2,401
	Thunder Springs	105	5,745	2,493	3,403
	Whitmore Wash	120	5,640	2,598	4,000
Temple Butte Formation		100	5,520	2,718	4,679
Tonto Group - Muav Formation		240	5,420	2,818	5,245
Tonto Group - Bright Angel Formation		NA	5,180	3,058	6,818

Notes:
1. As measured vertically from existing ground surface at bore entry location.
2. As measured along the bore path starting from the bore entry location.

Geologic formation thicknesses and contact elevations listed above and shown in Figure 5-1 were derived from literature review, analysis of site topography and surface weathering characteristics, and field observation. Variations between anticipated and encountered geologic formation thickness and true vertical contact depths along the bore path are estimated to be within ± 25 feet.

The bores will likely cross the Roaring Springs Fault within the Bright Angel Formation during directional drilling as shown in Figure 5-1. The offset of the fault is not large enough to cause a change in lithology, however, drill fluid and cement losses due to fracturing within the fault zone may occur while drilling within the Bright Angel Formation. The bores are not anticipated to encounter any large-scale faults prior to encountering the Bright Angel Formation.

5.3. Casing Pipe Design

Four telescoping steel casing pipes with diameters of 20-inches, 13 3/8-inches, 9 5/8-inches, and 7-inches respectively, will be installed within each bore. Casing pipe segments will be joined using threaded couplings as described in the Bore Drilling Plan (GKS, 2026). Installation of the casing pipes should account for the bore stability considerations for each formation as described in Section 4.2. Additional details regarding the casing pipe design are provided in the Bore Drilling Plan (GKS, 2026).

5.4. Drill Bit Considerations

Drill bits used to complete the bores are assumed to include Tricone and Polycrystalline Diamond Compact (PDC) bits. The contractor should account for the abrasivity of the rock type when choosing drill bits. Some rock types will be abrasive and cause excessive bit wear during drilling as described in Section 4.3. Additionally, the contractor should account for the possibility of bit loss in the case of drill string failure. Additional details regarding bit selection and bottom hole assemblies are provided in the Bore Drilling Plan (GKS, 2026).

5.5. Cementing

Each casing will be cemented into place within each bore following drilling. Cement losses during cementing may occur where formations are highly fractured or where karst features are present which will require additional cement. Stage tools and packers should be considered as part of the casing string to facilitate the cement operation. Table 5-2 provides values for cement take as a function of the degree of fracturing in each formation with respect to the theoretical annulus volume for the drill casing (i.e., 1.5 = 1.5 times the theoretical annulus volume). Additional details regarding the cement plan are included in the Bore Drilling Plan (GKS, 2026).

Table 5-2. Estimated Cement Take by Formation

Formation	Member	Cement Take[1]
Kaibab Formation	Fossil Mountain	2 – 3
Toroweap Formation	Woods Ranch	1 – 2
	Brady Canyon	1.5 – 2.5
	Seligman	1.5 – 2.5
Coconino Sandstone		1 – 2
Hermit Formation		1 – 2
Supai Group - Esplanade Sandstone		1 – 2
Supai Group - Wescogame Formation		1 – 2
Supai Group - Manakacha Formation		1 – 2
Supai Group - Watahomigi Formation		1 – 2
Redwall Limestone	Horseshoe Mesa	2 – 3
	Mooney Falls	2 – 3
	Thunder Springs	2 – 3
	Whitmore Wash	2 – 3
Temple Butte Formation		1.5 – 2.5
Tonto Group - Muav Formation		2 – 3
Tonto Group - Bright Angel Formation		1 – 2

Notes:
1. Drill casing theoretical annulus volume = 1.

GEOLOGIC UNITS:

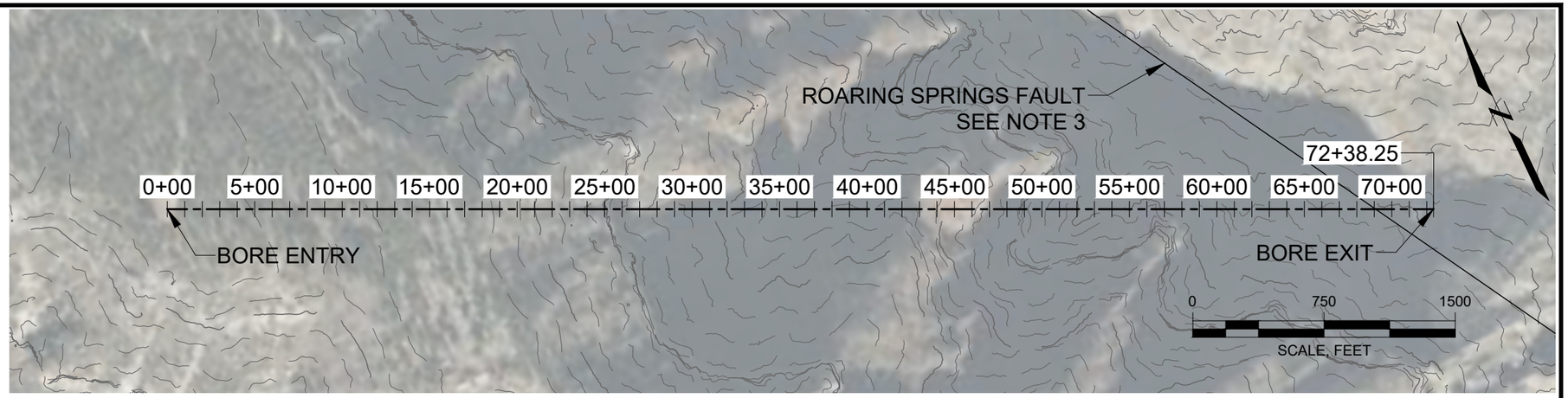
- KAIBAB FORMATION - FOSSIL MOUNTAIN MEMBER
- TOROWEAP FORMATION - WOODS RANCH MEMBER
- TOROWEAP FORMATION - BRADY CANYON MEMBER
- TOROWEAP FORMATION - SELIGMAN MEMBER
- COCONINO SANDSTONE
- HERMIT FORMATION
- SUPAI GROUP - ESPLANADE SANDSTONE
- SUPAI GROUP - WESCOGAME FORMATION
- SUPAI GROUP - MANAKACHA FORMATION
- SUPAI GROUP - WATAHOMIGI FORMATION
- REDWALL LIMESTONE - HORSESHOE MESA MEMBER
- REDWALL LIMESTONE - MOONEY FALLS MEMBER
- REDWALL LIMESTONE - THUNDER SPRINGS MEMBER
- REDWALL LIMESTONE - WHITMORE WASH MEMBER
- TEMPLE BUTTE FORMATION
- TONTO GROUP - MUAV FORMATION
- TONTO GROUP - BRIGHT ANGEL FORMATION

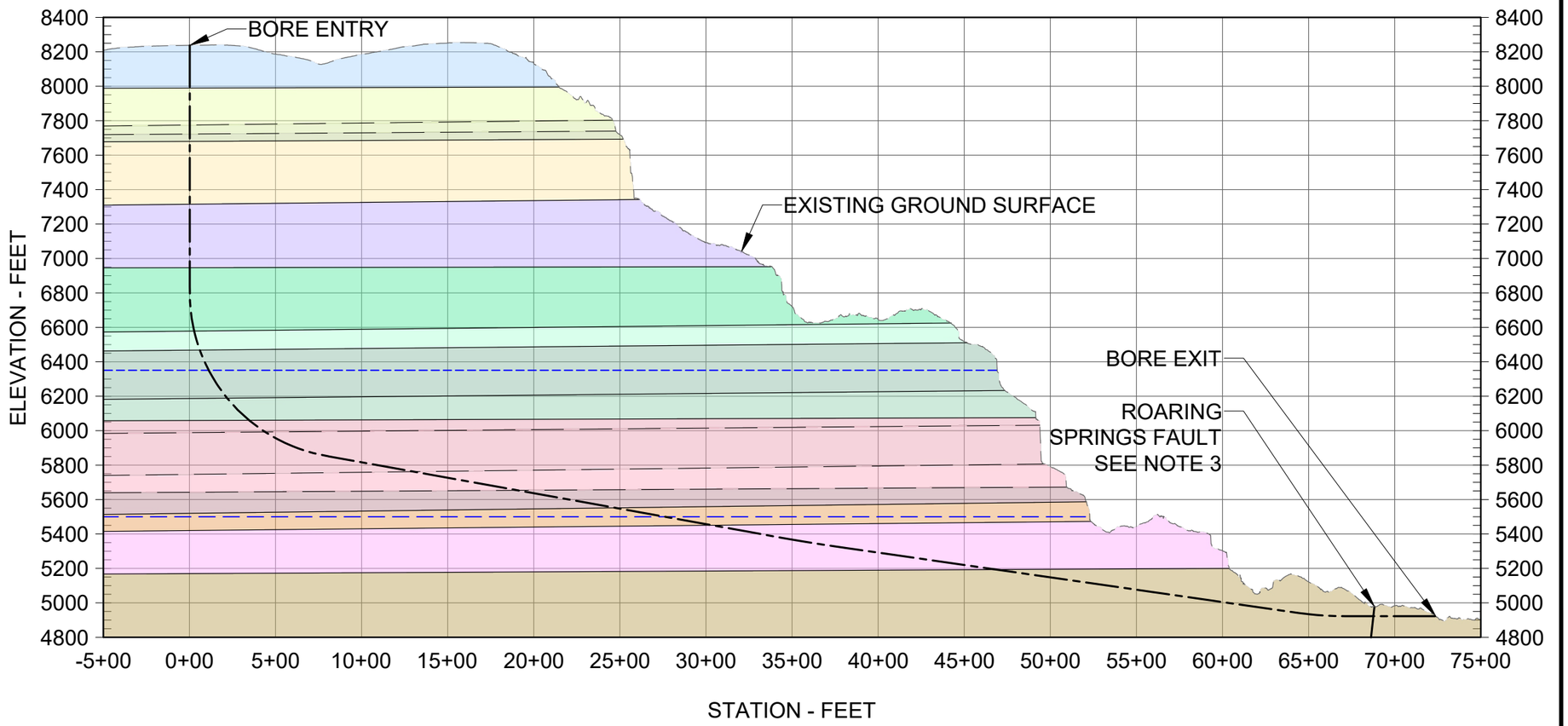
LINE TYPES AND SYMBOLS LEGEND:

- BORE ALIGNMENT
- FORMATION CONTACT ELEVATION
- MEMBER CONTACT ELEVATION
- C-AQUIFER APPROXIMATE ANTICIPATED GROUNDWATER ELEVATION
- R-AQUIFER APPROXIMATE ANTICIPATED GROUNDWATER ELEVATION

GENERAL NOTES:

1. NO VERTICAL EXAGGERATION SHOWN ON PROFILE.
2. ALIGNMENT AND INFRASTRUCTURE ARE AS SHOWN IN CONTRACT DRAWINGS AND PROVIDED HEREIN FOR INFORMATIONAL PURPOSES ONLY.
3. ROARING SPRINGS FAULT IS A NORTHWEST TO SOUTHEAST TRENDING REVERSE FAULT LOCATED WITHIN THE BRIGHT ANGEL FORMATION ALONG THE BORE ALIGNMENT. PRECISE LOCATION OF THE ROARING SPRINGS FAULT IS DIFFICULT TO LOCATE IN THE VICINITY OF THE BORE ALIGNMENT.





Geologic Interpretation Summary Report GRCA 318719 Rehabilitate North Rim Utility Infrastructure Grand Canyon North Rim, Arizona	GEI Consultants	INTERPRETED GEOLOGIC PROFILE
HDR Engineering, Inc. Fort Collins, Colorado	Project 2305987	July 2025 Fig. 5-1

6. Limitations

This study was conducted in accordance with generally accepted geotechnical engineering and engineering geologic practices and principles; no warranty, express or implied is made. The subsurface conditions described in this report were based on visual classification of widely spaced rock samples that may not be representative of in situ conditions, geotechnical laboratory testing on samples taken from the ground surface and performed by others, geophysical imaging, published literature for the general vicinity of the project site, information provided by the client, engineering judgement, and our experience with similar subsurface conditions and projects.

This report has been prepared exclusively for our client for design, bidding, and construction planning for the subject project. GEI is not responsible for technical interpretations by others of the data presented in this report or use of this report by others for the subject project or other projects. The interpreted geology and estimated values presented in this report are only intended as supplemental information for the proposed design and construction as understood by GEI at the time of issuing this report.

An environmental assessment was not included in GEI's scope of work for this project. Any statements regarding the absence or presence of hazardous and/or toxic substances presented herein are only intended for informational purposes. If the client is concerned about the environmental conditions at the site, GEI recommends the client and/or owner retain a qualified environmental firm to conduct an environmental site assessment.

7. References

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Appendix A Tunnelman's Ground Classification Chart

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Tunnelman's Ground Classification Chart (Rock)

Classification	Behavior	Additional Characteristics
Intact Rock	On account of the injury to the rock due to blasting, spalls may drop off roof several hours or days after blasting. This is known as spalling condition. Hard, intact rocks may also be encountered in the popping condition involving the spontaneous and violent detachment of rock slabs from sides or roofs.	Contains neither joints nor hair cracks. Hence, if it breaks, it breaks across sound rock.
Stratified Rock	In stratified rock, the spalling condition is quite common.	Consists of individual strata with little or no resistance against separation along the boundaries between strata. The strata may or may not be weakened by transverse joints.
Moderately Jointed Rock	Both the spalling and the popping condition may be encountered in rocks of this type.	Contains joints and hair cracks, but the blocks between the joints are locally grown together or so intimately interlocked that vertical walls do not require lateral support.
Blocky and Seamy Rock	Vertical walls may require support.	Consists of chemically intact or almost intact rock fragments that are entirely separated from each other and imperfectly interlocked.
Crushed Rock	If most or all of the fragments are as small as fine sand grains and no recementation has taken place, crushed rock below the water table exhibits the properties of water-bearing sand.	Crushed but chemically intact rock has the character of a crusher run.
Squeezing Rock	Rock slowly advances into the tunnel without perceptible volume increase.	Prerequisite for squeeze is a high percentage of microscopic and sub-microscopic particles of micaceous minerals or of clay minerals with a low swelling capacity.
Swelling Rock	Rock advances into the tunnel chiefly on account of expansion	The capacity for rock to swell is limited to those rocks that contain clay minerals such as montmorillonite, with a high swelling capacity.

(Adapted from Terzaghi, 1946)